



AI-Powered Data Analytics Internship

Week 1 Submission:

Data Understanding Report

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Executive Summary

This report presents the Week 1 deliverables of the internship project focused on predicting student participation in RIT learning opportunities, comprising a data dictionary, a cleaned dataset, and this summary statistics report. The historical dataset originally contained 8,558 raw learner signup records spanning 23 unique opportunities across five categories, namely Internship, Course, Event, Competition, and Engagement; after removing 15 test and QA records and correcting corrupted timestamp fields, the cleaned dataset retained 8,543 records ready for exploratory analysis and feature engineering in Week 2. The dataset reflects a genuinely international learner base, drawing from 1,813 unique institutions and 71 countries, with an overall participation rate of 47.57 percent across started, team allocated, and rewards awarded outcomes. Internships accounted for the largest share of activity, representing 5,417 records and 63.4 percent of all signups, while the median learner age at sign-up was 23.2 years, with an interquartile range of 21.7 to 25.2 years. Historical trends showed steady growth through 2023, with the strongest activity concentrated around mid-year and again at the start of 2024, while category-level analysis revealed that although Internships dominate in popularity, they are also the most competitive, with a substantially higher rejection rate than other categories. Together, these findings establish a reliable and well-documented foundation for the deeper exploratory analysis and modeling work planned for the following weeks of the internship.

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1. Introduction and Objectives

Artificial intelligence and generative AI are among the most important technologies of this generation. Organizations are relying more on data-driven systems to understand and predict human behavior, including how students engage with learning opportunities. This project aims to create a predictive model that estimates the likelihood of student participation in courses, internships, competitions, events, and activities offered through the RIT platform. The analysis in this report is based on the Student Participation Dataset provided for this project, which comes from the RIT Opportunity Wise Data file. Week 1 lays the foundation for that modeling work. The specific objectives were to:

- Examine the structure and scope of the historical student signup and engagement dataset
- Produce a data dictionary documenting every feature, its type, and its meaning
- Perform initial data cleaning, handling missing values, inconsistent formatting, and structural anomalies
- Generate summary statistics to surface early participation patterns

The three deliverables produced this week, the data dictionary, the cleaned dataset, and this summary statistics report, together form the reference foundation for Week 2's exploratory analysis and feature engineering.

2. Dataset Overview

Purpose of the Dataset

The dataset for this project tracks learner participation in career development and educational opportunities on the RIT platform. Each row details one learner's engagement with a specific opportunity. It includes demographic information, academic background, opportunity details, and application status. The goal of the dataset is to help the project team examine how learners find, apply for, and advance through the available opportunities. It also aims to identify trends in engagement over time, across different locations, and by opportunity type.

Dataset Scale and Coverage

The raw dataset includes 8,558 records across 16 variables. Each record represents a learner's sign-up for an opportunity. Therefore, a learner who signed up for multiple opportunities shows up more than once in the data. The records cover opportunity activity and learner signups, mainly recorded from January 2023 to March 2024. The final month in this range is a partial period rather than a complete one. This limitation is discussed in the historical analysis section of this report.

The dataset includes three main groups of information. The first group contains learner details like name, date of birth, gender, country, institution, and intended field of study. The second group contains opportunity details such as the opportunity name, category, and start and end dates. The third group includes participation details like the date a learner applied, their current status, and when their entry was created in the system.

What the Dataset Represents

Taken as a whole, the dataset reflects how a genuinely international group of young professionals engaged with structured learning and career opportunities. Learners in the data come from a wide range of countries, including the United States, India, Nigeria, Pakistan, Ghana, Kenya, Nepal, Egypt, and Bangladesh, among others. The opportunities themselves take several forms, including internships, courses, competitions, events, and engagement activities, each attracting a different profile of applicant and a different pace of progression, both of which are explored later in this report.

Key Observations

- The dataset combines demographic, academic, and engagement-related information within a single learner-level record.
- Participation is tracked through a clear timeline of signup, application, and opportunity dates, together with a status description and status code.
- Learners and opportunities are drawn from a broad and international base, which supports meaningful geographic and category-level comparisons.
- The structure of the dataset supports analysis of enrolment patterns, category performance, learner demographics, and application behaviour.

3. Data Dictionary

The table below describes each of the variables in the cleaned dataset with an explanation of what each field records, the data type it was recorded as, and why it is relevant to the analysis that is performed in subsequent sections of this report.

Variable Name	Data Type	Description	Analytical Relevance
Learner SignUp DateTime	DateTime	Date and time the learner signed up on the platform.	Enables tracking of enrolment activity and analysis of signup trends over time.
Opportunity Id	String (ID)	Unique identifier for the opportunity the learner engaged with.	Used to link and group participation records to a specific opportunity.
Opportunity Name	String	Title of the opportunity.	Provides context for opportunity-level analysis and reporting.
Opportunity Category	String	The category the opportunity belongs to, such as Internship or Course.	Central to the category performance and comparison analysis.
Opportunity Start Date	DateTime	Date and time the opportunity officially started.	Used to assess opportunity duration and scheduling patterns.
Opportunity End Date	DateTime	Date and time the opportunity officially ended.	Used together with the start date to calculate opportunity length.

First Name	Text	Learner's first name.	Helps identify repeat applicants and supports duplicate checking.
Date of Birth	Date	Learner's date of birth.	Used to calculate learner age and analyse demographic patterns.
Gender	Text (categorical)	Learner's self-reported gender.	Supports demographic analysis of participation across categories.
Country	Text (categorical)	Learner's country of residence.	Supports geographic analysis of where engagement is strongest.
Institution Name	String	Name of the learner's school or university.	Used to study academic background and identify feeder institutions.
Current/Intended Major	String	Learner's field of study.	Used to analyse academic trends among applicants.
Entry Created At	DateTime	Timestamp when the record was created in the system.	Useful for auditing and confirming record creation timing.
Status Description	String	Current status of the learner's engagement with the opportunity.	Core field for the application progression and funnel analysis.
Status Code	Integer	Numeric code corresponding to the status description.	Used for coding and automated status-related calculations.

Apply Date	DateTime	Date and time the learner applied to the opportunity.	Used to measure the time gap between signup and application.
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4. Data Quality Assessment

Before any analysis took place, the dataset was reviewed and cleaned in Python using pandas, with the process carried out in Google Colab. The raw file contained 8,558 records across 16 columns, and every check described below was verified against the final cleaned file that was handed off for analysis and visualisation.

Duplicate Records

No fully duplicated rows, meaning rows identical across every column, were found in the raw dataset. A closer check that matched learners by first name, date of birth, and opportunity identifier flagged 122 rows where the same learner appeared more than once against the same opportunity. These rows were kept rather than removed, since repeat entries of this kind can reflect a legitimate change in status over time, such as a learner moving from Applied to Team Allocated, rather than genuine duplication.

Missing Values

Missing values were reviewed both before and after cleaning. In the raw file, Institution Name and Current or Intended Major were each missing in 5 records, while Opportunity Start Date was missing in 3,794 records. Once the date columns were properly converted to a true datetime format, several previously hidden missing values surfaced, since corrupted text entries that had been stored as invalid timestamps were correctly reclassified as missing. After this step, Opportunity Start Date showed 4,637 missing records, Opportunity End Date showed 1,262, Apply Date showed 285, and Learner SignUp DateTime showed 275.

The missing Opportunity Start Date values were not treated as an error to correct. Because they are concentrated almost entirely among records with a Rejected or Waitlisted status, the pattern reflects opportunities that were never scheduled for learners who did not proceed, so the values were left as is to preserve that signal. The 5 missing Institution Name values were filled with the label Unknown so that no records were lost, while Current or Intended Major was left blank where no reliable value could be inferred.

Inconsistencies

All six date and time columns in the dataset were originally stored as plain text and were converted to a proper datetime type, with corrupted entries such as extra digits or invalid hour values coerced to missing during conversion. Two logical consistency checks were then applied. The first checked for opportunities whose end date fell before their start date, which flagged 527 rows, all traced back to just three opportunities where the start and end dates had been entered in reverse. These were corrected by swapping the two dates for the affected opportunities, resolving every flagged row. The second check, which looked for application dates falling after an opportunity had already ended, found no affected records.

Text formatting was also reviewed for casing, spacing, and naming inconsistencies. First Name was standardised to title case, extra whitespace was stripped, and entries containing numbers or symbols were replaced with Unknown. Institution Name required the most corrections of any field in the dataset. Saint Louis University alone appeared under roughly 25 to 30 different spellings in the raw data, including variants such as Saint Louise University, Saint Louis University, St Louis University, and SLU. A manually built lookup table was used to consolidate these variants into one canonical name, while genuinely distinct institutions, such as St. Louis Senior High School, were deliberately kept separate. This work reduced the number of unique institution values from 2,089 to 1,746 and consolidated 4,218 records under the single canonical name Saint Louis University.

Summary of Cleaning Checks

Check Performed	Result	Action Taken
Exact duplicate rows across all columns	0 found	None required.
The same learner applying to the same opportunity more than once	122 rows flagged	Retained, since repeat entries can reflect legitimate status changes rather than true duplication.
Missing Opportunity Start Date	4,637 records after date typing	Left as is, since the gap is concentrated in Rejected or Waitlisted records and

		reflects a structural pattern rather than an error.
Missing Institution Name	5 records	Filled with the label <i>Unknown</i> so that no records were lost.
Opportunity End Date earlier than Opportunity Start Date	527 rows flagged, traced to 3 opportunities	Start and end dates were swapped for the affected opportunities, resolving all flagged rows.
Institution name spelling variants	2,089 unique values reduced to 1,746	Standardised through a manually built lookup table while keeping genuinely distinct institutions separate.

Potential Issues and Limitations

- A small number of low-frequency institution name variants, together representing fewer than 20 records, remain unmerged. Their combined share of the dataset is small enough that they are not expected to materially affect institution-level results.
- March 2024 contains a noticeably lower signup count than surrounding months. This is judged to reflect an incomplete final month of data collection rather than a genuine decline in learner interest, and it is treated as a limitation rather than a finding in the historical analysis that follows.
- 2024 figures throughout this report cover only the first quarter of the year, so year-over-year totals for 2023 and 2024 are not directly comparable and should be read with that in mind.
- No records were removed from the dataset during cleaning. The final cleaned file retains all 8,558 rows and 16 columns, with corrected data types, standardised text fields, and documented missing values.

5. Historical Analysis

This section looks at how learner signups moved from month to month and year to year across the period covered by the dataset, highlighting when applications were strongest, when they slowed, and what those patterns suggest.

Overall Application Trends

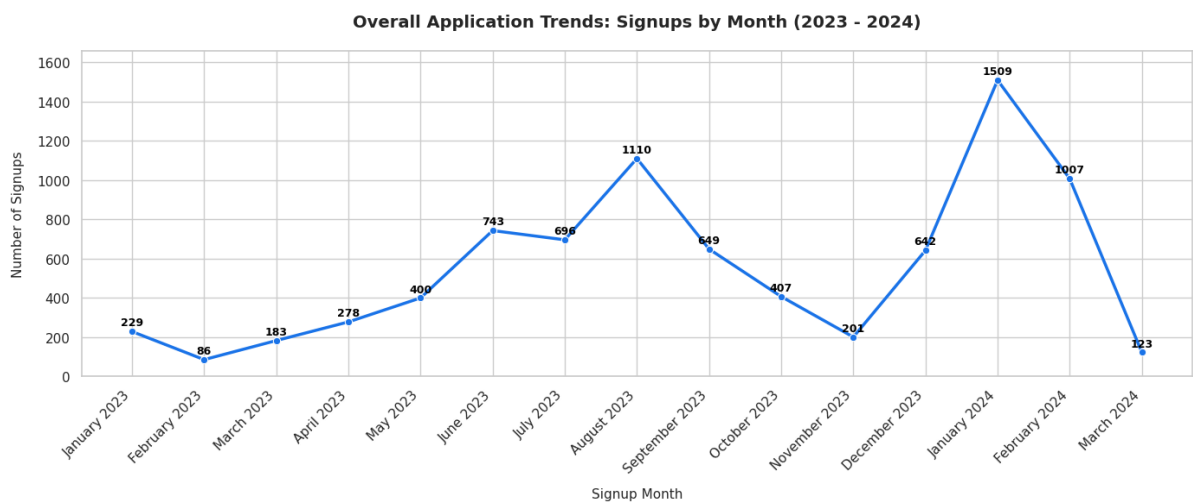


Figure 1. Overall monthly signup trend, 2023 to 2024.

Learner signups grew steadily across the period covered by the dataset. Activity was modest at the start, rising from 229 signups in January 2023 to 400 by May, before climbing sharply between June and August 2023, when monthly signups reached 743, 696, and a peak of 1,110, respectively, the busiest stretch of the year. Signups then eased from September through November, falling to 201 by November, before recovering to 642 in December as the year closed. January 2024 recorded the highest signup count anywhere in the dataset, at 1,509, followed by a still strong 1,007 in February. March 2024 recorded only 123 signups, a figure best explained by an incomplete final month of data rather than a genuine drop in interest, and it should not be compared directly against the other months.

Growth and Decline Patterns

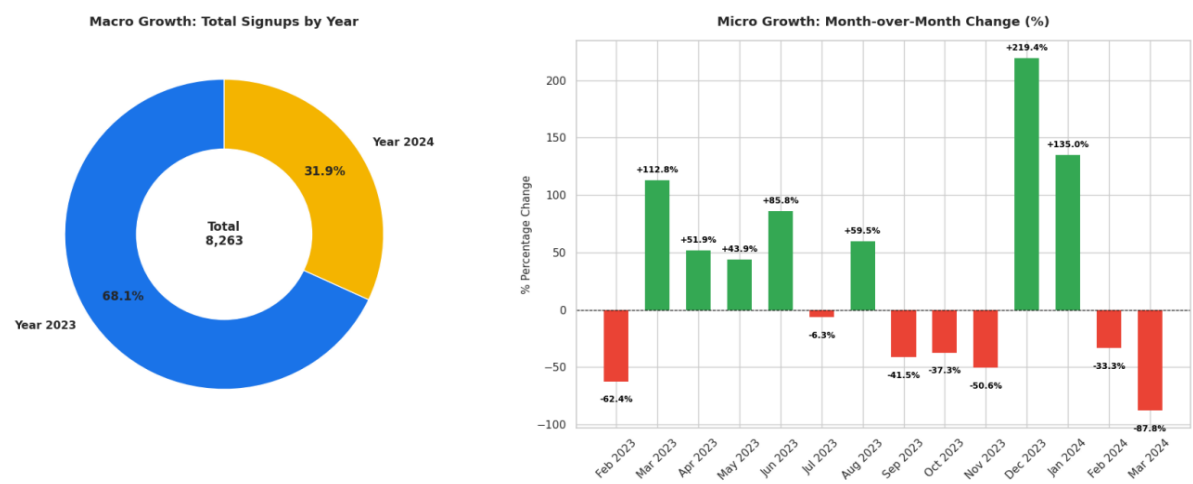


Figure 2. Yearly signup share and month-over-month percentage change.

A total of 5,624 learners signed up in 2023, compared with 2,639 in 2024, though the 2024 figure covers only the first three months of the year and is not a fair full-year comparison. Month-over-month growth moved in clear swings rather than a straight line. Signups fell 62.45 percent in February 2023 before rising consistently from March through June, peaking at plus 85.75 percent in June. A small dip of 6.33 percent in July was followed by a 59.48 percent rise in August, before signups declined in every month from September to November. December brought a sharp recovery of plus 219.40 percent, and January 2024 extended that momentum with growth of plus 135.05 percent, reaching the highest signup level in the dataset. Growth turned negative again in February 2024, down 33.27 percent, and fell further in March, down 87.79 percent, again most likely reflecting incomplete data for that month.

Significant Observations

Learner interest did not move at a constant pace but followed a repeating cycle, with the strongest growth occurring in December and January and the sharpest declines occurring in February and, due to incomplete records, March of the following year. Because the 2024 figures cover only the first quarter of the year, they should not be judged against the full-year total for 2023. The overall pattern suggests that the middle of the year and the start of a new year are the periods that consistently attract the highest learner interest, and this rhythm is worth monitoring on a year-over-year basis rather than being judged from any single month.

In summary, learner interest rises and falls in a repeating pattern, with December and January standing out as the strongest months across the dataset.

6. Opportunity Category Analysis

This section examines how the five opportunity categories in the dataset, namely Internship, Course, Competition, Engagement, and Event, compare in terms of popularity, variety, and how far learners progressed after applying.

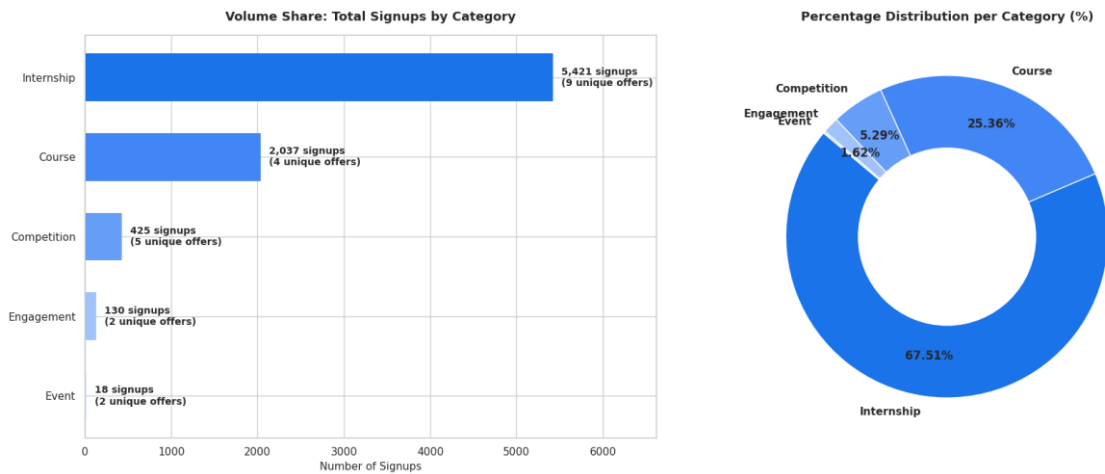


Figure 3. Signup volume and percentage share by opportunity category.

Highest Performing Categories

Internships were the clear leader in the dataset, receiving 5,421 signups and accounting for roughly two out of every three learners overall. Courses were the second most popular choice, drawing 2,037 signups and making up close to one quarter of all activity. Internships also offered the widest range of listings, with 9 unique opportunities, showing that both demand and choice were greatest in this category.

Lowest Performing Categories

Engagement opportunities received the fewest signups of any category, at just 130, making it the weakest performer by demand. Events and Competitions also trailed well behind Internships and Courses, with 545 and 425 signups respectively. Engagement additionally offered the smallest number of unique listings, with only 2 opportunities available, meaning both interest and choice were limited here.

Category Distribution

Internships hold the largest share of all signups at 67.50 percent, meaning roughly two out of every three learners applied for an internship. Courses make up the next largest share at 25.36 percent, close to one in every four learners. Competitions account for 5.29 percent, Engagement for 1.62 percent, and Events for just 0.22 percent, together forming a small remaining slice of total activity. Internships and Courses combined therefore account for approximately 93 percent of every signup recorded in the dataset.

Application Progression by Category

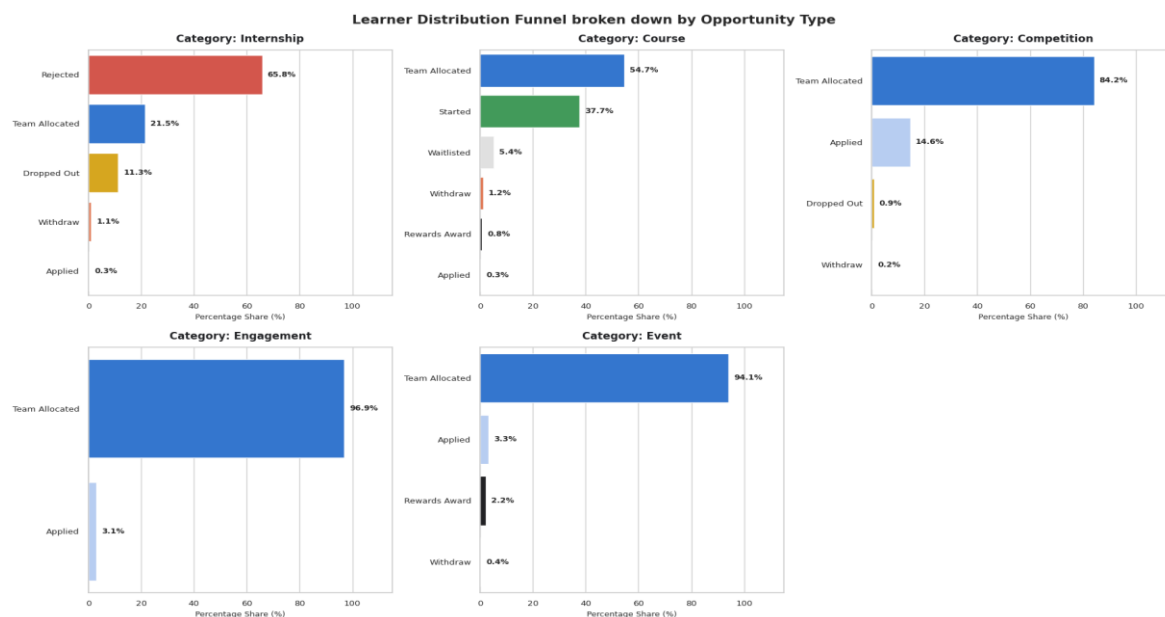


Figure 4. Status funnel breakdown for each opportunity category.

Beyond raw signup counts, it is useful to see how far learners actually progressed after applying. Competition applicants progressed well, with 84.24 percent reaching the Team Allocated stage. Course applicants also progressed strongly, with 54.69 percent reaching Team Allocated and 37.65 percent shown as having started the course. Engagement had the smoothest progression of any category, with 96.92 percent of learners reaching Team Allocated, and Events performed similarly well at 94.13 percent. Internships followed a markedly different pattern. Despite being the most popular category by far, 65.84 percent of internship applicants were rejected, and only 21.49 percent reached the Team Allocated stage, with a further 11.31 percent dropping out along the way.

Popularity and ease of entry are not the same thing. Most categories accept the large majority of their applicants, but Internships reject about two out of every three people who apply, making them the most competitive category in the dataset by a clear margin.

7. Additional Insights

Beyond the core historical and category trends covered above, further summary visuals were produced to explore who the learners were, where they came from, what they studied, and how quickly they moved from signing up to applying. These graphs add useful context to the headline figures already presented.

Learner Demographics

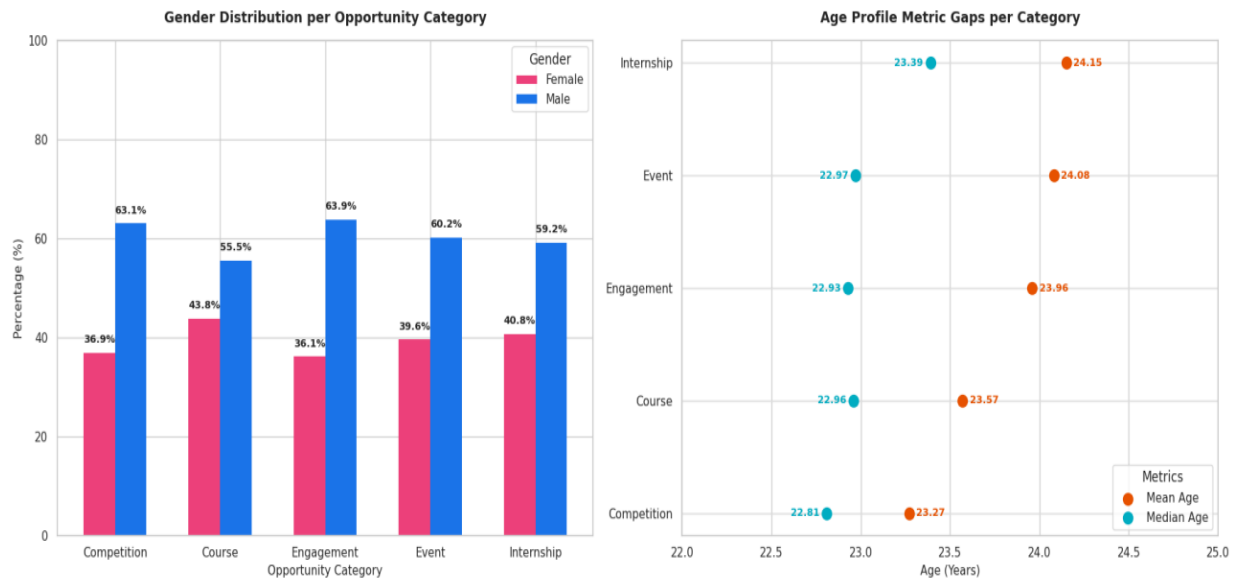


Figure 5. Gender distribution and age profile by opportunity category.

Male learners formed the majority in every category, while female participation ranged from roughly 37 to 44 percent, with Courses showing the highest share of female learners at 43.84 percent. Competitions and Engagement had the highest proportion of male learners, both above 63 percent. The average age of learners was between 23 and 24 years across all categories, with Internship applicants recording the highest average age at 24.15 years and Competition participants the youngest at 23.27 years. Overall, the learner base is mostly male but with meaningful female representation, and is consistently made up of young adults in their early twenties regardless of category.

Geographic Reach

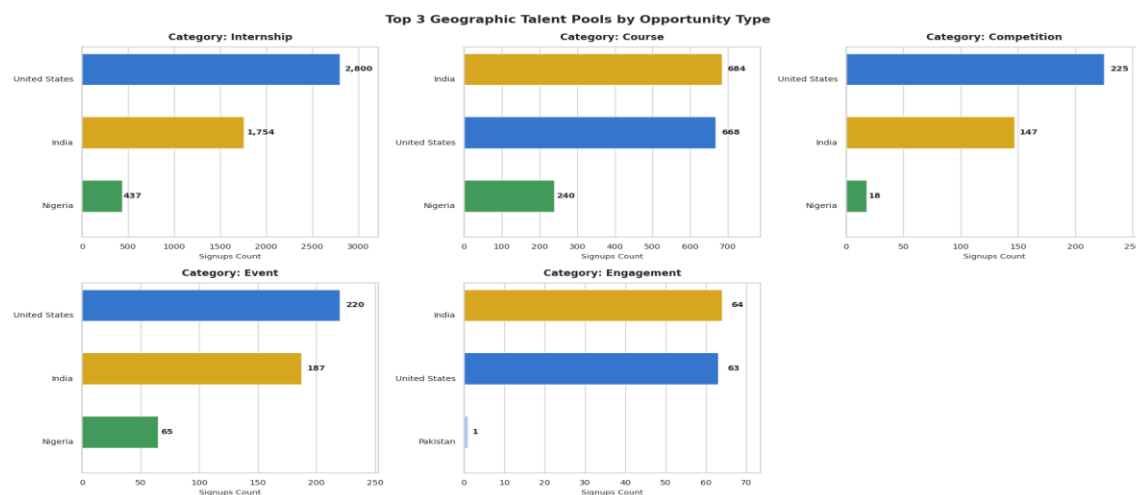


Figure 6. Top three contributing countries by opportunity category.

The United States and India were the two leading sources of learners across almost every category. The United States led participation in Internships, Competitions, and Events, while India led participation in Courses and Engagement activities. Nigeria consistently appeared as a third significant contributor across most categories. This spread suggests the opportunities have reached a genuinely international learner base, with the United States, India, and Nigeria forming the core of overall engagement.

Academic Background

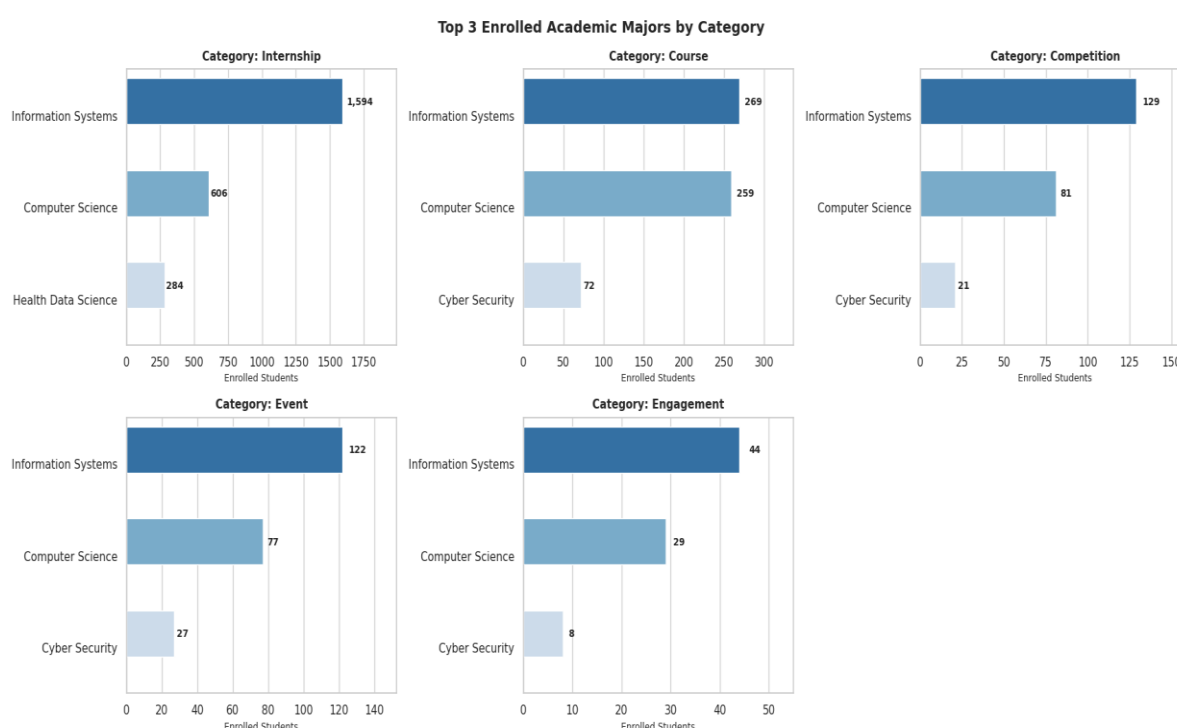


Figure 7. Top majors by opportunity category.

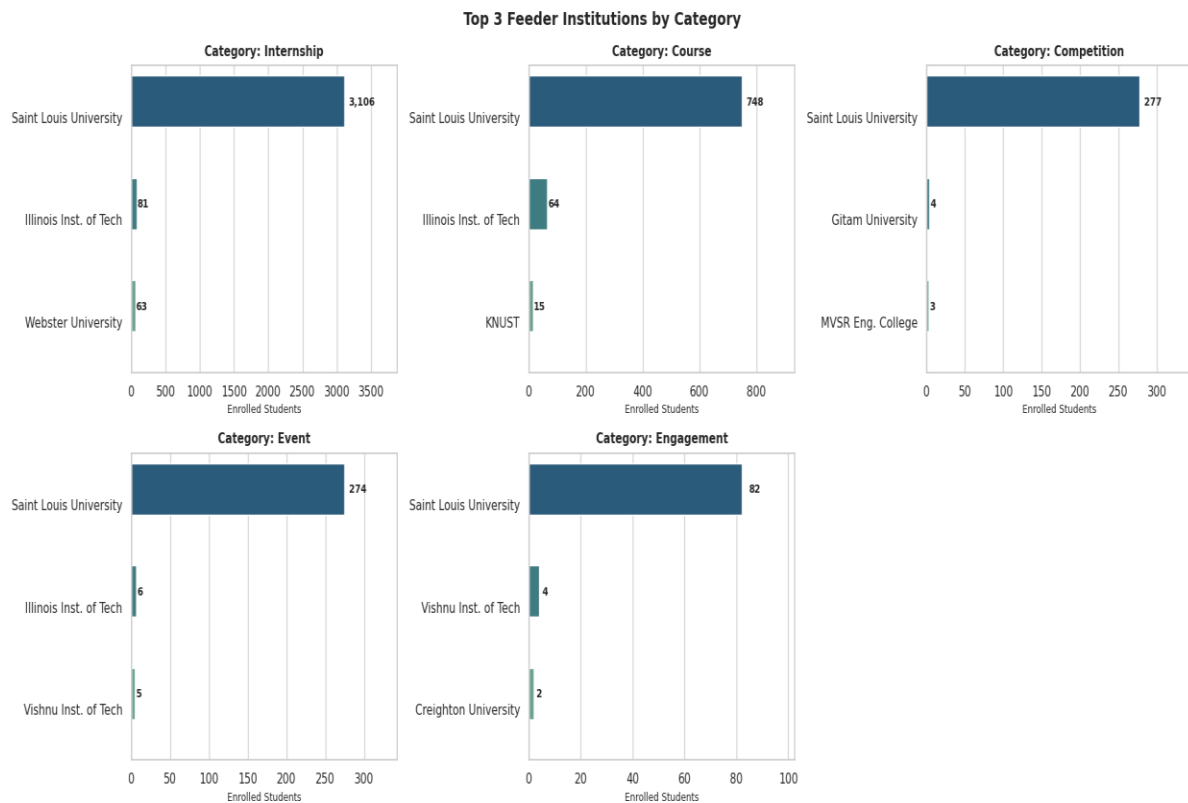


Figure 8. Top feeder institutions by opportunity category.

Information Systems was the most common major in every opportunity category, with Computer Science consistently ranking second. The third most common major varied by category, with Cyber Security appearing for Courses, Competitions, Events, and Engagement, and Health Data Science taking third place for Internships. Saint Louis University was the largest single source of learners in every category by a wide margin, with Illinois Institute of Technology the next largest contributor for Courses, Internships, and Events. This points to a learner base concentrated around technology and data-related fields of study, anchored heavily around one partner institution alongside a wider group of smaller contributing schools.

Application Behaviour

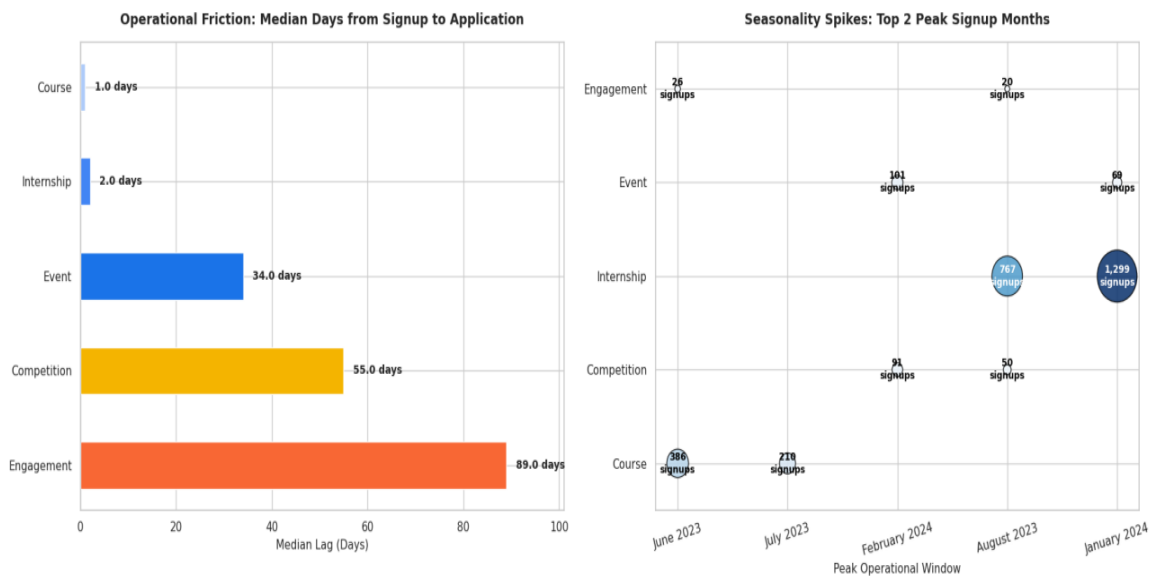


Figure 9. Median days from signup to application, and peak signup months by category.

Courses had the shortest gap between signup and application, with learners applying within a median of just 1 day, and Internships were almost as fast, at a median of 2 days. Events took a long time to complete, with a median time of 34 days; competitions, with a median gap of 55 days; and Engagement activities, with the longest delay of all, at a median of 89 days. Courses and Engagement had its peak signup activity in June 2023. Internships had its peak signup activity in January 2024. Competitions and Events had its peak signup activity in February 2024. This indicates a clear behavioural split in that learners move almost immediately for Courses and Internships but take significantly longer to commit to Events, Competitions and Engagement opportunities; useful information when planning when to promote each type of opportunity.

8. AI-Assisted Analysis

Artificial intelligence tools were used to support parts of the analysis carried out for this project, from writing and refining code to simplifying the wording of the final interpretations presented in this report.

AI Tools Used

- Google Gemini, used through its Advanced Data Analysis feature, helped generate and improve the Python code used for the analysis and supported the overall data exploration process.
- ChatGPT was used to explain analysis results, draft simple interpretations, and improve the wording of the findings presented throughout this report.

Prompts Used

- Asked for suggestions to improve how the analysis was presented and visualised.
- Requested simple Python code that could be run directly in Google Colab.
- Asked for improvements to existing code so that outputs would be easier to understand.
- Requested simplified, report-friendly wording for the interpretation of each result.

Validation Process

- All AI-generated Python code was executed and tested directly in Google Colab before being relied upon.
- Outputs were manually checked against the underlying dataset to confirm that values and percentages were accurate.
- Final interpretations were reviewed to make sure they accurately reflected what the underlying analysis actually showed.

9. Conclusion

The four areas of work in this report, understanding the dataset, documenting the dataset structure, assessing and correcting the quality, and analysing the patterns within the dataset, all work together to give us a clear picture of how learners engage with the platform. The dataset is pretty good, with no missing records and only a few missing values and formatting issues, which we were able to fix without removing any records. When we look at the history, we can see that the number of learners grew slowly in 2023, with a lot of people using the platform in the middle of the year and at the start of 2024. We also looked at the categories and found that Internships are the most popular, but also the hardest to get into, while Courses, Events, Competitions, and Engagement opportunities are not as popular, but people who apply for them are more likely to succeed. The pictures and charts we made help us

understand the learners better, showing us that they are young, from around the world, and really into technology, and that they behave differently depending on what category they are in. All of these findings will help us with the rest of the internship project. The internship project will be based on these findings. The internship project is going to use these findings as a foundation for the internship project.

10. Additional Resources

Resource Type	Format	Description	Document Link / Path
Data Dictionary	Excel file	Detailed schema mapping out all variable names, data types, and analytical definitions.	File link: Click Here
Data-Cleaning Notebook	Jupyter Notebook	The complete workflow for handling duplicates, date validation, and text standardization.	File link: Click Here
Cleaned Dataset	Excel File	Cleaned dataset of 8,543 records with corrected data types, standardized fields, and resolved inconsistencies.	File link: Click Here
Analysis-Visualization Notebook	Jupyter Notebook	Exploratory data analysis, core metrics, descriptive statistics, and reporting visuals.	File link: Click Here